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Article

Paleomagnetism of the Grum Zn–Pb–Ag deposit, Selwyn Basin, Yukon, Canada

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Abstract

The Grum deposit is one of five large clastic-dominated Zn–Pb–Ag deposits in the Anvil district of the Selwyn Basin, Yukon. Grum's mineralization occurs at the contact between the late Proterozoic to Cambrian Mount Mye Formation and the Cambrian to Early Ordovician Vangorda Formation. Paleomagnetic analyses of 113 specimens from 15 mineralized sites in the Grum open pit were undertaken to date Grum's mineralization. The analyses isolated a stable characteristic remanent magnetization (ChRM), mostly by thermal and then alternating field demagnetization. The main ChRM carrier is single- or pseudosingle-domain pyrrhotite with minor magnetite that give a paleopole at 67.8°N latitude and 317.4°E longitude (radius of the cone of 95% confidence, $A_{95} = 7.2°$). The paleopole yields an age of 176 ± 12 Ma for the Grum deposit after a clockwise rotation and northward translation of the basin to best fit the North American apparent polar wander path. Thus, the Zn–Pb–Ag Grum mineralization significantly predates intrusion of the nearby mid-Cretaceous Anvil Batholith, but is coeval with widespread Early Jurassic metamorphism in the Selwyn Basin that correlates with the Early Jurassic collision of the Intermontane Belt terranes with the North America craton.

Key words: Paleomagnetism, remagnetization, Zn–Pb–Ag mineralization, Grum deposit, Yukon

Résumé

Le gisement Grum fait partie de cinq grands gisements de Zn–Pb–Ag à prédominance clastique dans le district d'Anvil du bassin de Selwyn (Yukon). La minéralisation à Grum se trouve au contact entre la Formation du mont Mye, d'âge protérozoïque tardif à cambrien, et la Formation de Vangorda, d'âge cambrien à ordovicien précoce. Des analyses paléomagnétiques de 113 spécimens provenant de 15 sites minéralisés dans la fosse à Grum ont été réalisées afin de déterminer l'âge de la minéralisation. Les analyses ont permis d'isoler une magnétisation rémanente caractéristique (MRC) stable, principalement par démagnétisation thermique, puis par champ alternatif. Le principal porteur de la MRC est de la pyrrhotite à domaine unique ou pseudo-unique avec un peu de magnétite, qui produisent un paléopôle de latitude 67,8°N et de longitude 317,4°E (rayon du cône de l'intervalle de confiance de 95 %, $A_{95} = 7{,}2°$). Le paléopôle donne un âge de 176±12 Ma pour le gisement Grum après une rotation horaire et une translation vers le nord du bassin, pour la meilleure concordance avec la trajectoire nord-américaine de dérive apparente des pôles. Ainsi, la minéralisation en Zn–Pb–Ag du gisement Grum est significativement plus ancienne que l'intrusion du batholithe d'Anvil, d'âge crétacé moyen, situé à proximité, mais est contemporaine du métamorphisme jurassique précoce répandu dans le bassin de Selwyn qui est corrélé avec la collision au Jurassique précoce des terranes du domaine intermontagneux avec le craton nord-américain. [Traduit par la Rédaction]

Mots-clés : paléomagnétisme, remagnétisation, minéralisation en Zn–P–Ag, gisement Grum, Yukon

1. Introduction

The Paleozoic Selwyn Basin in Yukon is dominated by marine sedimentary units containing black shales and cherts that host three substantial clastic-dominated (CD) Zn–Pb–Ag districts (Goodfellow and Lydon 2007), namely the (*i*) Anvil in upper Cambrian strata, (*ii*) Howards Pass in lower Silurian strata, and (*iii*) MacMillan Pass in Upper Devonian strata (Goodfellow 2007) (Fig. 1). The Anvil district includes five large Zn–Pb (±Ag, ±Ba) deposits, i.e., the Faro, Vangorda, Swim, Grum, and Dy deposits (Fig. 2). These five deposits have an estimated combined reserve of 120 Mt with a combined Pb + Zn grade of 9.3% (Jennings and Jilson 1986). The five deposits have been categorized as syngenetic or diagenetic and submarine exhalative in origin, and to be sedimentary exhalative (SEDEX) deposits (Pigage 1990). However, a hydrothermal epigenetic origin should not be excluded because these deposits have synsedimentary characteristics such as a well-developed hydrothermal compositional

Fig. 1. Major tectonic elements in the Northern Cordillera and location of the Anvil Zn–Pb–Ag District in the Selwyn Basin of Yukon, Canada. The box with dashed outline indicates the study area. Modified from Creaser and Spence (2005). Distributions of major and minor Zn–Pb deposits are also plotted (Goodfellow 2007).

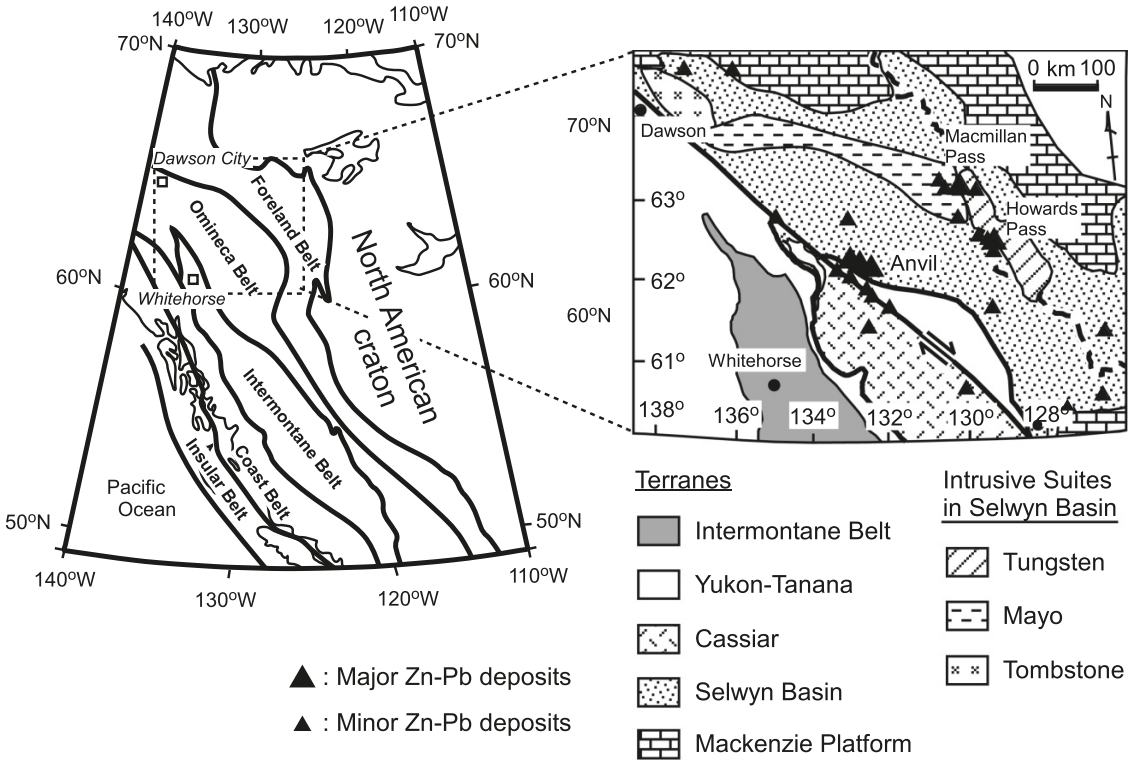
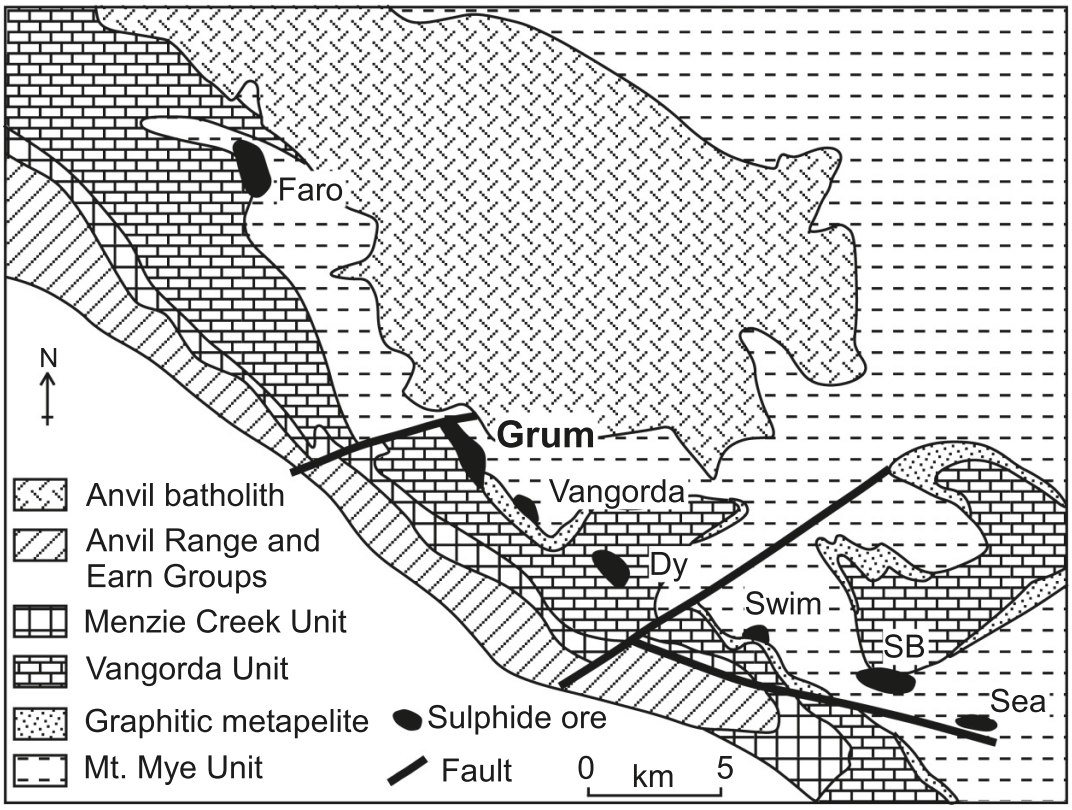
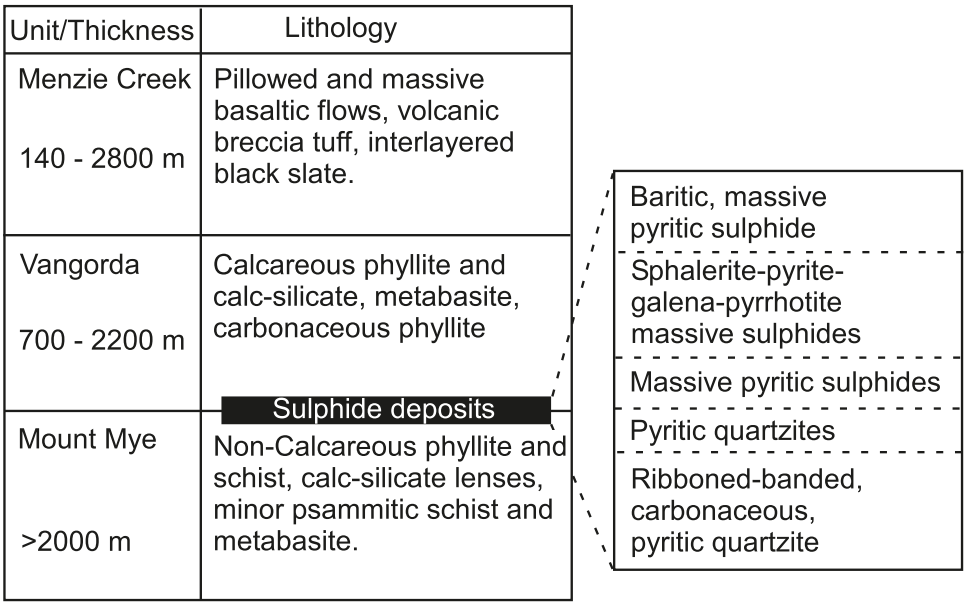


Fig. 2. Simplified geology of the Anvil district after Shanks et al. (1987) and Pigage (1990).



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Fig. 3. Late Proterozoic to Silurian stratigraphy of the Anvil District (modified from Pigage 2004; Cobbett 2014).



layering in most sulphide facies (Leach et al. 2005; Leach et al. 2010). Also, Goodfellow et al. (1995) noted a spatial association between the CD deposits and rift-related magmatism in the Selwyn Basin. The lack of precise radiometric age dating of the mineralization has created uncertainty as to the genesis of these CD deposits. Paleomagnetic methods have provided age constraints for the similar Howards Pass CD deposits (Kawasaki and Symons 2012). This paleomagnetic study of the Grum deposit in the Anvil district was undertaken to date Grum's mineralization and to see if it shared a temporal history with the Howards Pass CD deposits.

2. Geology

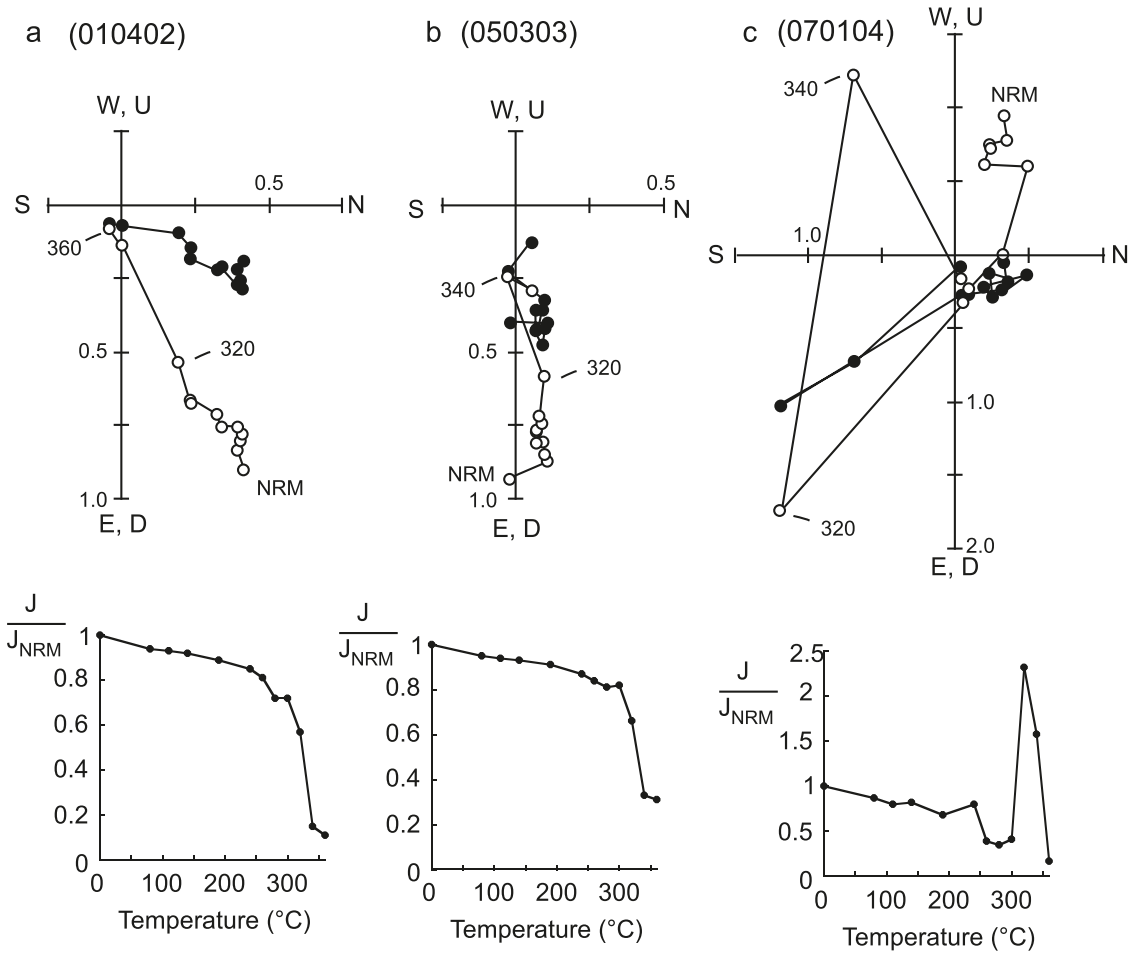
The Anvil district is located within the Selwyn Basin of the Omineca Belt in the northern Canadian Cordillera (Fig. 1). The basin has a complex history of extensional tectonism from late Proterozoic to Mississippian (Goodfellow 2007). The Anvil district on the basin's western side contains a structurally thickened sequence of late Precambrian to Permian metasedimentary and metavolcanic schists and phyllites (Jennings and Jilson 1986) that host five major CD deposits and that have been intruded by Cretaceous granites and granodiorites (Pigage and Anderson 1985) (Fig. 2). The pre-Silurian sequence is divided into three units: (*i*) the noncalcareous metapelitic Mount Mye Formation, (*ii*) the calcareous pelitic Vangorda Formation, and (*iii*) the basaltic Menzie Creek Formation (Jennings and Jilson 1986) (Fig. 3). The late Proterozoic to Cambrian Mount Mye Formation consists of brown, fine- to medium-grained biotite–muscovite schist (Cobbett 2014). The overlying Vangorda Formation consists of fine-grained calc-silicate schists and hornfels (Cobbett 2014), and it is lithologically similar to the Cambro-Ordovician Rabbitkettle Formation of the Road River Group, which indicates that the Vangorda Formation ranges in age from Cambrian to Early Ordovician (Pigage 1990). The basaltic Menzie Creek Formation conformably overlies the Vangorda Formation and yields fossils that indicate an Ordovician to Silurian age (Pigage 2004). The overlying Devonian to Permian sequence is divided into the Earn and Anvil Range Groups that are characterized by shale, chert, and basalt with minor limestone and coarse chert-rich clastic sediments (Tempelman-Kluit 1972; Gordey and Anderson 1993) and are unrelated to the ore deposits (Pigage 1990).

Grum's Zn–Pb–Ag mineralization occurs at the Mt. Mye – Vangorda contact, forming an approximately 150–200 m thick stratigraphic interval. The interval contains (*i*) graphitic phyllite with trace sulphides, (*ii*) massive sulphides with variable amounts of barite, and (*iii*) siliceous sericitic phyllite with bands, lens, and blebs of sulphide (Bulatovic and Wyslouzil 1999). Jennings and Jilson (1986) inferred that the Anvil deposits were aligned with the edge of a reduced sedimentary facies in a second-order extensional basin, and therefore, they suggested a submarine synsedimentary exhalative genetic model for the mineralization although no direct evidence for a synsedimentary origin is preserved.

Five deformation events, D_1 to D_5, have been recognized in the Anvil district. The D_1 event is related to northeast folding, thrusting and nappe emplacement during the pre- to mid-Cretaceous (Jennings and Jilson 1986). D_2 is related to the emplacement, uplift, and unroofing of the Anvil Batholith that caused southwest-directed folding and greenschist to amphibolite facies metamorphism of the adjacent strata (Jennings and Jilson 1986; Smith and Erdmer 1990). The Anvil Batholith is composed of porphyritic granodiorite and is a member of the Selwyn plutonic suite (Hart et al. 2004). The batholith and surrounding schist have given K–Ar cooling ages between 81 and 104 Ma (Wanless et al. 1967; Tempelman-Kluit 1972; Wood and Armstrong 1982) and a whole rock multimineral Rb–Sr age of 100 ± 4 Ma (Pigage and Anderson 1985). Heffernan and Mortensen (2000) determined a U–Pb zircon

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Fig. 4. Thermal step demagnetization of example specimens from sites 1, 5, and 7 that have initial natural remanent magnetization (NRM) intensities (J_{NRM}) of 1.35×10^{-4}, 9.68×10^{-4}, and 2.86×10^{-5} A/m, respectively. Solid (open) circles denote projection on the horizontal (vertical) plane defined by the north (N), east (E), south (S), and west (W) ((north (N), down (D), south (S), and up (U)) axes. Axial intensity values (J) are expressed as a ratio of the NRM intensity. Some steps are labeled in °C.



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age of 106.4 ± 0.4 Ma for the Mt. Billings Batholith, which is also a member of the Selwyn plutonic suite. The metamorphic grade decreases outward from the Anvil Batholith. Thus, although the adjacent Faro deposit is metamorphosed to the amphibolite facies, the Grum deposit is only metamorphosed to the lowermost greenschist facies. The minor D_3, D_4, and D_5 events produced only minor folding and steeply-dipping crenulation foliations (Brown and McClay 1993).

3. Methods and results

3.1. Sampling and initial measurements

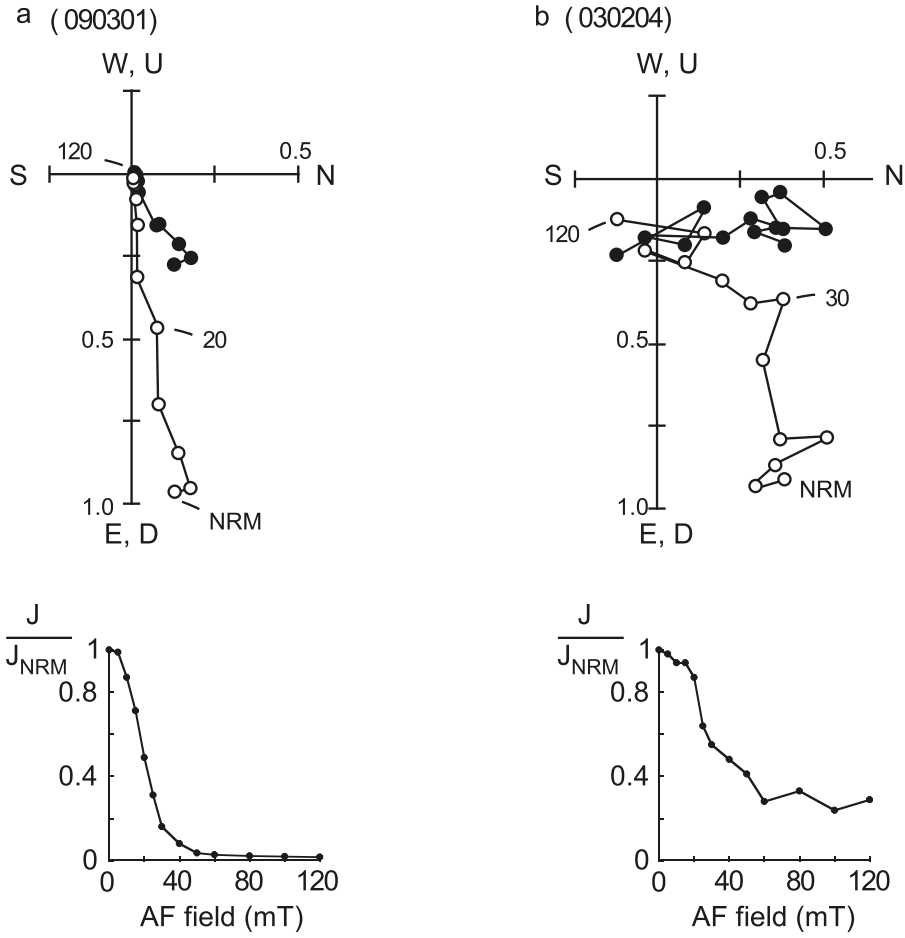
Three or four oriented mineralized block samples were collected at each of 15 sites in the Grum mine pit. Five to ten core specimens, each 2.54 cm in diameter and ~2.20 cm long, were prepared for each site at the University of Toyama. All 113 specimens were stored for several weeks at the University of Windsor in a shielded room with an ambient field of ~100 nT to allow their unstable and undesirable viscous remanent magnetization (VRM) components to substantially decay. Subsequent magnetic measurements were done in the shielded room. The natural remanent magnetizations (NRM) of the specimens were measured on a 2G Enterprises 755R DC-SQUID magnetometer with a specimen sensitivity limit of ~2×10^{-6} A/m. The median NRM intensity for the 113 specimens was 1.20×10^{-4} A/m (quartiles: $Q_1 = 4.15 \times 10^{-5}$; $Q_3 = 6.96 \times 10^{-4}$).

3.2. Demagnetization

One test specimen from each site was thermally demagnetized in 11 steps up to 360 °C using a Magnetic Measurements MMTD-80 oven with the steps biased into the diagnostic unblocking temperature ranges of goethite (80–140 °C) and pyrrhotite (260–340 °C) (Dunlop and Özdemir 1997). A second test specimen from each site was alternating field (AF) demagnetized in 11 steps up to 120 mT using a Sapphire Instruments SI-4 demagnetizer.

About one-third of the thermal test specimens show slow decreases or slight increases in remanence intensity up to ~260 °C and then a sharp remanent intensity drop between ~280 °C and ~340 °C (Fig. 4*a*). About half of the thermal

Fig. 5. Alternating field (AF) step demagnetization of example specimens from sites 3 and 9 with initial NRM intensities (J_{NRM}) of 8.94 × 10^{-4} and 3.76 × 10^{-3} A/m, respectively. Plotting conventions as in Fig. 4. Some steps are labeled in mT.



test specimens show either similar slow decreases or slight increases up to ~280 °C and then a moderate remanence intensity drop between ~300 °C and ~360 °C, leaving more than 20% in NRM intensity at 360 °C (Fig. 4*b*). The remaining thermal test specimens show erratic intensity increases and directional changes up to 300 °C (Fig. 4*c*).

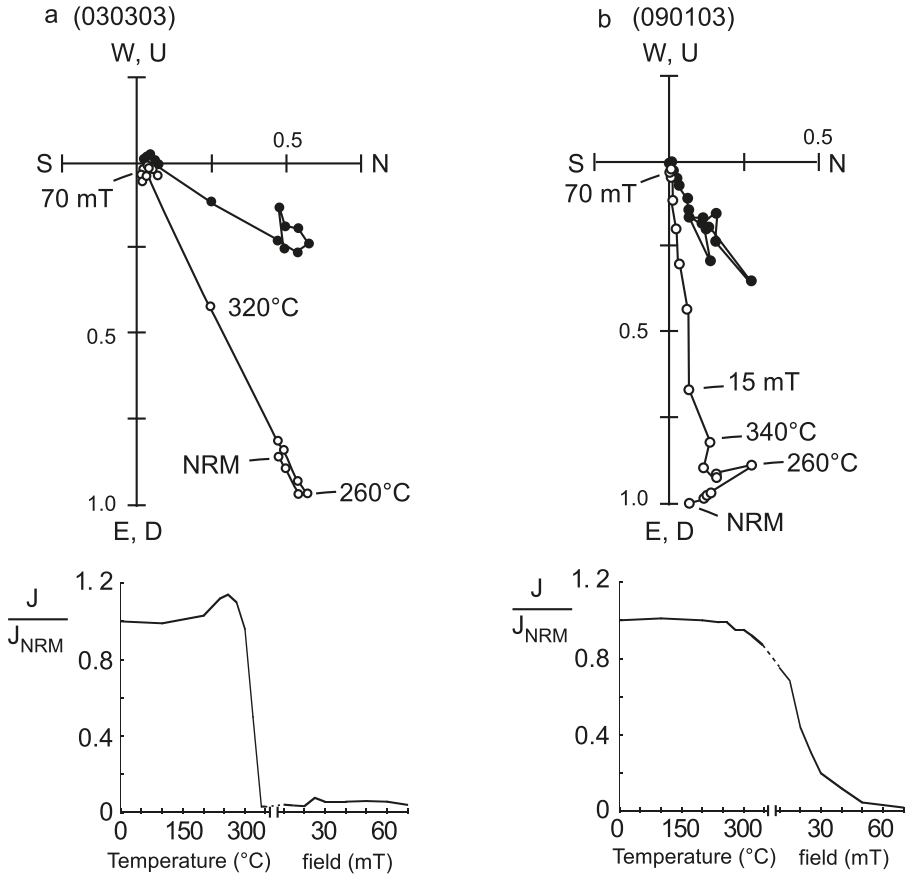
About one-third of the AF test specimens show rapid remanence intensity decreases up to ~50 mT, followed by moderate remanence intensity decreases up to ~120 mT (Fig. 5*a*), whereas the rest of the specimens (except for sites 1 and 2) show continuous moderate remanence intensity decreases during AF demagnetization (Fig. 5*b*). Unfortunately, sulphide minerals such as pyrite oxidize to form new magnetite in the oven during the heating process. Also, pyrrhotite is one of the main remanence carriers and it has a high coercivity that cannot be entirely AF demagnetized by the ~150 mT maximum field of the AF demagnetizer. As a compromise, the remaining 83 specimens were thermally demagnetized in eight steps up to 340 °C and then AF demagnetized in 10 steps up to 100 mT (Figs. 6*a* and 6*b*).

The specimen remanence directions were plotted by demagnetization step on Schmidt equal-area nets to define the directional density contours (Kamb 1959) that showed the NRM is directed moderately–steeply downward to the northeast. The NRM direction migrates to a slightly steeper inclination and rotates slightly clockwise in declination up to about 240 °C and remains constant thereafter on subsequent thermal and AF demagnetization (Fig. 7). Thus the Kamb (1959) analysis shows that the remanence is very stable and virtually unaltered since its emplacement. The specimen characteristic remanent magnetization (ChRM) directions were determined traditionally using a combination of orthogonal vector component plots (Zijderveld 1967), principal component analysis (Kirschvink 1980), and remagnetization circle analysis (Bailey and Halls 1984). Site and unit mean directions were calculated following Fisher (1953) (Fig. 8; Table 1). The Grum samples give a mean ChRM direction for the mineralization of declination (D) = 40.5°, inclination (I) = 70.9° (number of sites (N) = 15, radius of cone of 95% confidence (α_{95}) = 4.6°, and precision parameter of Fisher (1953) (k) = 69.6). The ChRM direction is significantly different at 95% confidence from the present Earth's magnetic field direction at Grum (average 1914–2013 is D = 26.6°, I = 77.8°; International Geomagnetic

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Fig. 6. Thermal and then AF step demagnetization of example specimens from sites 3 and 9 with initial NRM intensities (J_{NRM}) of 3.18 × 10^{-3} and 3.93 × 10^{-3} A/m, respectively. Plotting conventions as in Fig. 4. Some steps are labeled in °C or mT.



Reference Field (IGRF-13)), which indicates that the ChRM records a major primary or secondary geological event.

3.3. Rock magnetics

The magnetic mineralogy and their effective domain size were tested using stepwise isothermal remanent magnetization (IRM) acquisition analysis on rock fragments from sites 4, 5, and 10. A Lake Shore Cryotronics PMC MicroMag model 2900 alternating gradient magnetometer (AGM) was used to magnetize each sample in a direct current (DC) field up to 1000 mT. A value of 0.9 × J_{1000} is used here to define "saturation", where J_{1000} is the remanence intensity at 1000 mT. The moderate IRM acquisition to saturation by about 200 mT indicates that the major mineralization's magnetic minerals are pseudosingle domain (PSD) to single domain (SD) pyrrhotite or SD magnetite (Fig. 9) (Symons and Cioppa 2000). Note that the relatively noisy data are caused by the very weak magnetic intensities of the samples.

4. Discussion

4.1. Location of Grum paleopole

A sharp remanent intensity drop between ~280 and ~340 °C during thermal demagnetization with erratic intensity increases and directional changes up to 300 °C indicate that both hexagonal and monoclinic pyrrhotite are present in the specimens (Dunlop and Özdemir 1997). In addition, the moderate IRM acquisition to saturation during the stepwise IRM acquisition tests shows that the main remanence carriers in the Grum mineralization are SD and (or) PSD pyrrhotite with minor magnetite. The mean ChRM direction for the Grum Zn–Pb–Ag mineralization gives a pole position at 67.8°N, 317.4°E (radius of the cone of 95% confidence A_{95} = 7.2°) (Fig. 10). This paleopole, like the paleopole for the Howards Pass Zn–Pb–Ag mineralization, falls off the present Mesozoic apparent polar wander path (APWP) for North America (Torsvik et al. 2012). Paleomagnetic studies of the 97 Ma Ragged Pluton (Symons et al. 2008) and the 98 Ma Cantung W–Cu ore deposit (Kawasaki and Symons 2014) on the northeastern side of the Selwyn Basin (Fig. 1) have given concordant paleopoles to North America, indicating the basin has been autochthonous, at least since the Cretaceous. Also, there is abundant evidence that the Selwyn Basin strata originated as a marginal slope and basin assemblage of the Canadian miogeoclinal shelf (e.g., Gordey and Anderson 1993).

4.2. Implications for the tectonic history of the Selwyn Basin

The deviations of the Grum and Howard Pass CD paleopoles relative to the North American APWP imply that the Selwyn

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Fig. 7. Directional density contours (Kamb 1959) for the specimen ChRM directions from the Grum deposit on thermal and then AF step demagnetization on segments of equal-area projections of the lower hemisphere. The initial contour is at the 5σ level. Upper-hemisphere remanence directions are reversed to their antipodal direction.

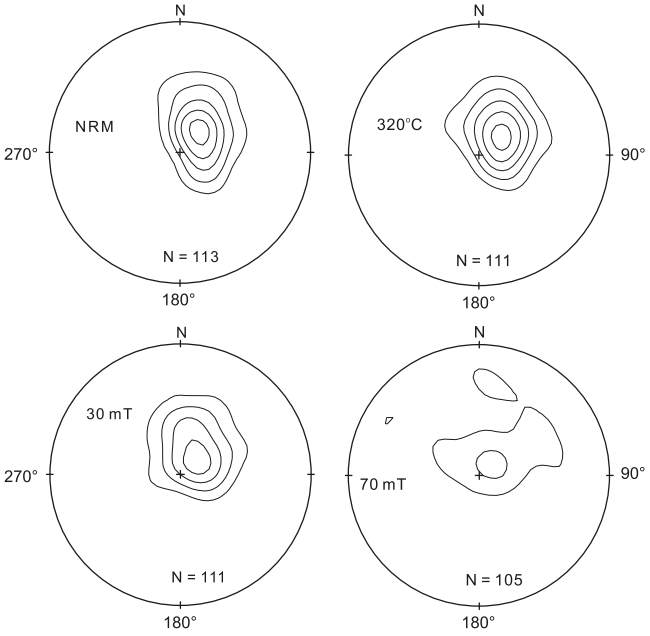
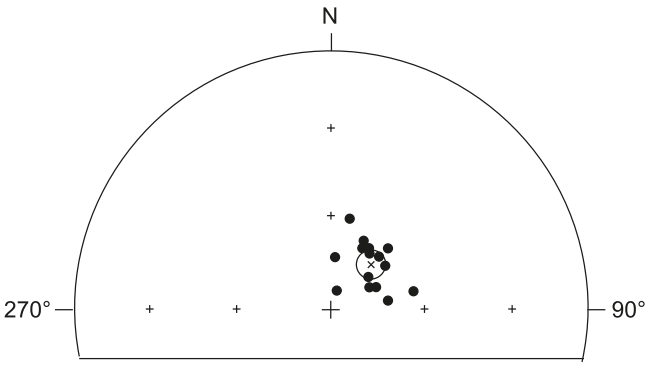


Fig. 8. Site mean ChRM directions for the Grum collection where X is the overall unit-mean ChRM direction, surrounded by its cone of 95% confidence (Fisher 1953).



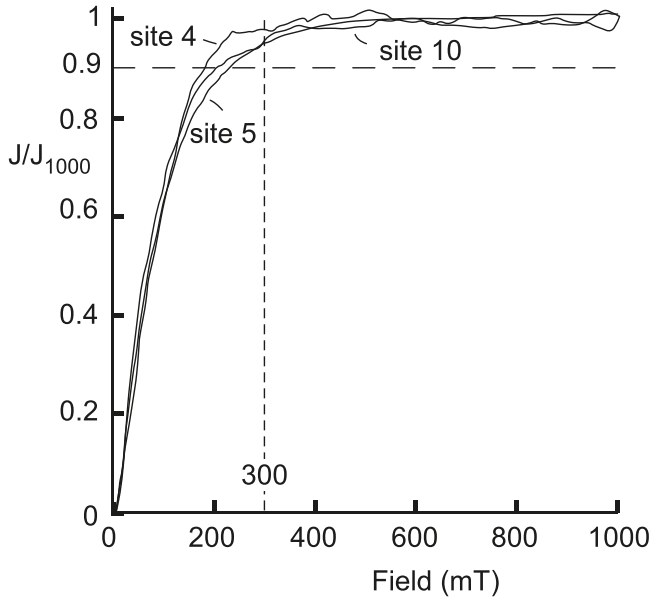
Basin underwent tectonic motion relative to North America prior to the mid-Cretaceous. Following Demarest (1983), a paleomagnetic age of 176 ± 12 Ma (2σ) is determined for the Grum deposit from a clockwise rotation of 31° ± 9° and a northward translation of 13° ± 7° to best fit the North American APWP (reference paleopole at 176 Ma: 72.5°N, 119.9°E, A_{95} = 4.0°; Torsvik et al. 2012). The paleopole for the Howards Pass CD deposits was compared to the North American APWP of Besse and Courtillot (2002) by Kawasaki and Symons (2012). Here we compare it to the updated North American APWP

Table 1. Site mean characteristic remanent magnetization (ChRM) directions.

Site	N^E	N^R	Dec. (°)	Inc. (°)	α_{95} (°)	k
1	2	3	3.5	73.1	13.5	42.3
2	6	0	26.4	68.0	11.0	38.3
3	7	0	31.2	66.9	9.9	38.1
4	8	0	25.0	65.7	10.4	29.5
5	6	0	77.1	63.2	9.5	51.1
6	4	1	33.6	68.3	13.4	35.3
7	4	2	42.4	63.3	14.2	31.6
8	3	2	47.8	74.2	19.9	17.8
9	8	0	80.8	71.8	8.3	45.6
10	4	3	41.3	67.4	13.0	67.4
11	4	3	50.5	67.9	8.4	57.7
12	3	2	15.2	83.7	21.1	16.0
13	8	0	58.8	76.0	8.9	39.3
14	8	0	62.5	74.2	8.3	46.0
15	8	0	11.4	60.2	8.1	47.5
Unit mean direction			40.5	70.9	4.6	69.6

Note: Mean declination (Dec.), inclination (Inc.), and radius of cone of 95% confidence (α_{95}) in degrees and precision parameter (k) of Fisher (1953). N^E, specimen end points; N^R, remagnetiz remagnetization circles.

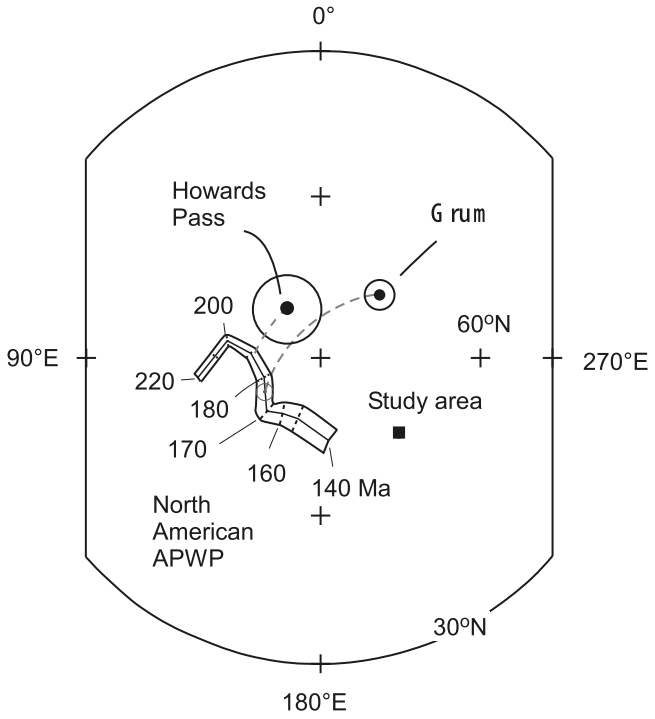
Fig. 9. Stepwise isothermal remanent magnetization acquisition curves for three rock fragment samples taken from sites 4, 5, and 10.



of Torsvik et al. (2012), which gives an age of 189 ± 22 Ma after a clockwise rotation and northward translation of 23° ± 19° and 0° ± 13°, respectively (reference paleopole at 189 Ma: 71.9°N, 88.8°E, A_{95} = 3.0°; Torsvik et al. 2012). Taken together, the Grum and Howards Pass CD Zn–Pb–Ag paleomagnetic results, which come from the outboard and inboard sides of the Selwyn Basin, respectively, imply (*i*) that the basin is

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Fig. 10. Segment of an equal-area polar projection, centered on 90°N for the Earth, showing the Grum deposit's measured paleopole relative to: the 220–140 Ma North American apparent polar wander path (APWP) of Torsvik et al. (2012). Also shown is the paleopole for the Howards Pass Zn–Pb–Ag mineralization.



para-autochthonous, (*ii*) that it was deposited on North America's western margin, and (*iii*) that the basin moved as a microplate during the Early Jurassic collision of the Intermontane Belt (IMB) terrane against the North American margin. The Grum and Howards Pass paleomagnetic estimates come from opposite sides of the basin and are not significantly different. The estimates' average suggest that the basin was translated about 700 ± 700 km northward along the North American margin and was rotated about 27° ± 14° clockwise during translation. The northward translation and clockwise rotation estimates are consistent in sense but smaller in magnitude than the values recorded in the outboard Yukon-Tanana terrane and IMB (Symons et al. 2005; Symons and Kawasaki 2019).

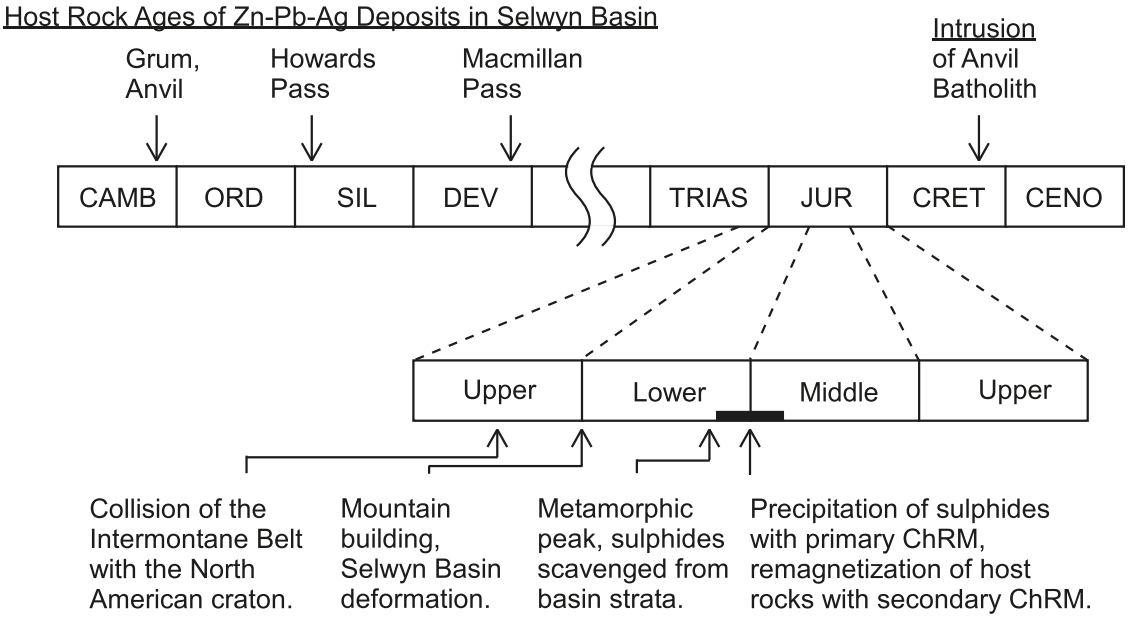
Greenwood et al. (1991) reported that the most intense metamorphism recorded in the Omineca Belt occurred in the early Middle Jurassic, after the Early Jurassic collision of the IMB terranes against North America and during their subsequent obduction. The average age of 182 ± 7 Ma for the Grum and Howards Pass Zn–Pb–Ag mineralization agrees well with the early Middle Jurassic estimate for maximum metamorphic reheating in the Selwyn Basin. Further, this reheating event would have been sufficient to cause remobilization of the primary fine-grained sulphides and magnetite to form coarse-grained and remagnetized pyrrhotite and magnetite with their present ChRM direction.

4.3. Implications for CD deposit genesis

Five Zn–Pb–Ag CD deposits in the Anvil district, including Grum, occur along a southwesterly curvilinear trend away from the 100 ± 4 Ma Anvil Batholith (Fig. 2). Goodfellow et al. (1995) noted a spatial association between the Zn–Pb–Ag CD districts and rift-related magmatism in the Selwyn Basin. Based on whole rock geochemical analyses of samples from the Anvil, Keno Hill, and MacMillan Pass CD deposit districts, Leybourne et al. (2018) found that the alkalic rocks in the Selwyn Basin have high Ba and Tl contents, which implies that these rocks have either a direct magmatic origin or that they underwent hydrothermal alteration. Hence they suggest the possibility of a genetic link between the hydrothermal fluids and the magmatic rocks, rather than simple diagenesis of a primary marine deposit. The paleomagnetic age of ~182 Ma for the Grum deposit precludes the possibility of a genetic link between the Grum deposit (and the other four CD deposits in the Anvil district) and the Anvil Batholith (Fig. 11).

The similar 182 ± 7 Ma ages for the widely-separated Grum and Howards Pass Zn–Pb–Ag CD deposits favour a simple hydrothermal model for their ore genesis because it requires only one major tectonic event, late Early to early Middle Jurassic orogenesis, to drive ascending hydrothermal mineralizing fluids into multiple widely-separated sites for sulphide precipitation. The fluids would find such sites in strata containing either intergranular void spaces or, more likely, replaceable primary mineral grains. Hydrothermal fluids are widely produced by rainfall on mountain highlands that sink through the crust, progressively acquiring geothermal heat and dissolved sulphide ions. At depth, the hydrostatic pressure gradient drives the fluids outward beneath adjacent lowland basins and back towards surface through any available faults where they cool and precipitate metallic sulphides in any suitable deposit sites. In contrast, syngenetic, diagenetic and submarine exhalative models of ore genesis are complex. They require, improbably, the Selwyn Basin to have been dissected by three major tectonic rift events at ~490 Ma, again at ~430 Ma, and then again at ~360 Ma to emplace the primary fine-grained metallic mineralization of the Anvil, Howards Pass, and MacMillan Pass CD deposits, respectively. Also, active rift zones rapidly push both sides of the dissected crust apart by tens to hundreds of kilometers, and they infill the gap with volcanic rocks. The Selwyn Basin is far too small and shows no evidence for three rift events from ~490 Ma to ~360 Ma. Further, the three rejected models of ore genesis require the Grum and Howards Pass CD ore deposits to have undergone chlorite facies metamorphism at ~187 Ma to explain their paleopoles. A metamorphic event to over ~300 °C is required to remobilize the hydrothermal system and convert the fine-grained sulphides to coarse grained and to re-align the pyrrhotite and magnetite magnetic domains from a primary Paleozoic to an ambient Jurassic ChRM direction (Dunlop and Özdemir 1997). Paleomagnetic or radiometric dating of the MacMillan Pass and other Selwyn Basin CD deposits would provide additional tests of the favoured hydrothermal genetic hypothesis for these deposits.

Fig. 11. Schematic overview for the formation of the Grum and other clastic-dominated (CD) Zn–Pb–Ag deposits in the Selwn Basin, Yukon. CAMB, Cambrian; ORD, Ordovician; SIL, Silurian; DEV, Devonian; TRIAS, Triassic; JUR, Jurassic; CRET, Cretaceous; CENO, Cenozoic.



5. Conclusion

Based on step demagnetization and rock magnetic tests, the main remanence carriers in the Grum Zn–Pb–Ag mineralization are SD and (or) PSD pyrrhotite with minor magnetite. The Grum paleopole at 67.8°N, 317.4°E ($A_{95} = 7.2°$) is significantly different at 95% confidence from the North American APWP. The best paleomagnetic and geologic fit for the Grum deposit to the North American APWP is given by a clockwise rotation of 31° ± 9° and a northward translation of 13° ± 7°, resulting in a paleomagnetic age of 176 ± 12 Ma for the Grum deposit. The 176 ± 12 Ma paleomagnetic age is significantly older than the mid-Cretaceous Anvil Batholith, providing a strong evidence that there is no genetic link between the formation of the Grum Zn–Pb–Ag deposit and intrusion of the Anvil Batholith. Combined with paleomagnetic results from the Howards Pass Zn–Pb–Ag deposits on the other side of the Selwyn Basin, the results indicate a late-Early to early-Middle (182 ± 7 Ma) Jurassic remobilization event took place in the Paleozoic strata of the Selwyn Basin following the Early Jurassic collision of the IMB terrane against North America. The results of this study favour a hydrothermal model of ore genesis at ~182 Ma for the Grum and other CD Zn–Pb–Ag deposits in the Selwyn Basin.

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Data availability

The data that support the findings of this study are available from the corresponding author, KK, upon reasonable request.

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Competing interests

The authors declare that they have no competing interests.

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